Structural Modelling, Validation, and Analysis of *HeV-g2* Attachment Glycoprotein

In-Silico Characterisation using SWISS-MODEL, ProSA-web, and UCSF Chimaera

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Introduction & Objective

- *HeV-g2* is a newly discovered *Henipavirus* variant recently identified in horses and bats.
- The experimental 3D structure of its attachment glycoprotein is currently unknown, creating a gap in our understanding of the virus.
- This project generates a 3D homology model for the protein *(Accession: UCY33688)* to analyse its structural stability compared to the pathogenic Hendra virus.
- These insights are critical for predicting how the virus interacts with host receptors.

Methodology

6pdl.4.A Attachment glycoprotein

Crystal Structure of Hendra Virus Attachment G Glycoprotein in Complex with Receptor Ephrin-B2

	0.75	-	92.82	X-ray, 2.7Å	hetero-dimer ⚠	1 x NAG-NAG-BMA, 1 x NAG-NAG, 1 x NAG-NAG-BMA-MAN-MAN-FUC-FUC, 4 x NAG

- Selected the *HeV-g2* attachment glycoprotein (Accession: UCY33688) to serve as the target sequence.
- Identified the *Hendra Virus* glycoprotein (PDB ID: 6PDL) as the optimal structural template.
- Achieved a 92.82% sequence identity match, as indicated by the blue alignment bar in the results.
- Ensured high model confidence, since identity scores above 90% typically yield accurate structures with minimal error.

Stereochemical Validation

- Assessed the stereochemical quality of the protein backbone using the Ramachandran plot via PROCHECK

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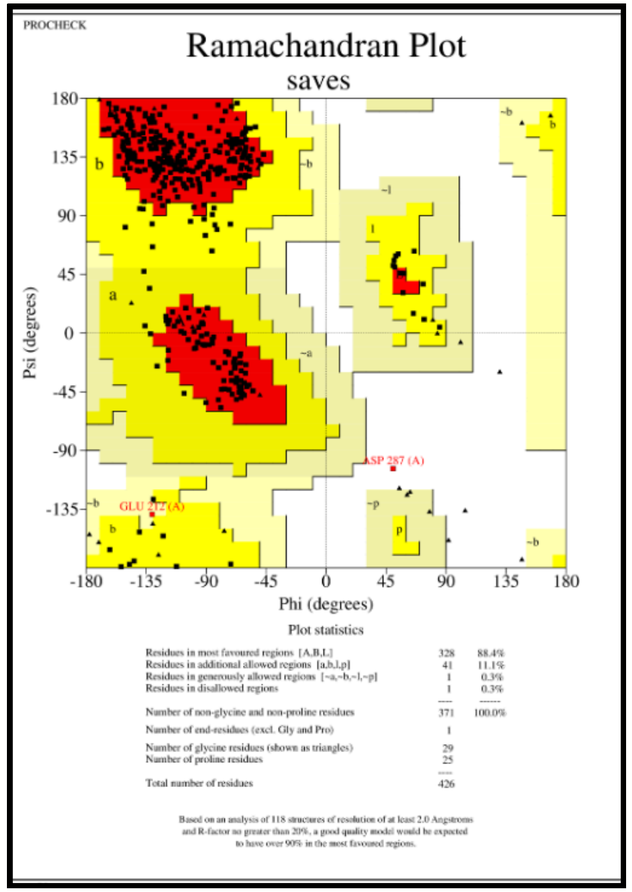
- Observed that 88.4% of residues fall within the "Most Favoured Regions" (red areas), indicating a highly stable structure.

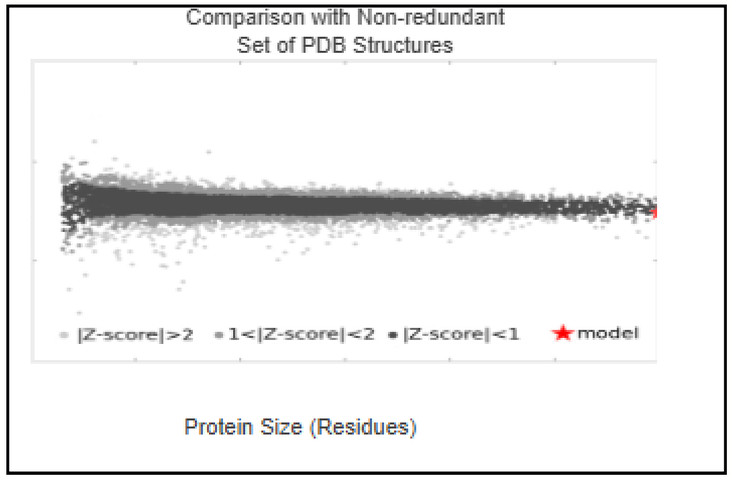
- Outperformed the reference study, which reported a slightly lower score of 86.9% in favoured regions.

- Confirmed excellent model quality, with only 0.3% outliers (a single residue) found in the disallowed regions.



Global Quality Estimation

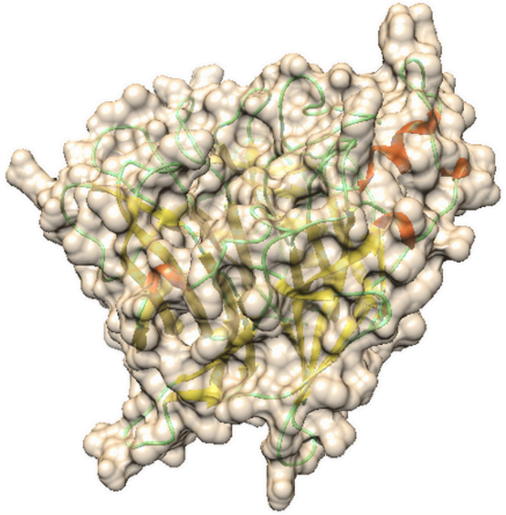
- Evaluated the model's energy profile against a database of experimentally determined protein structures (X-ray and NMR).
- Achieved a Z-Score of -7.91, indicating the model adopts a stable, native-like conformation.
- Surpassed the reference study's benchmark, which reported a Z-score of -7.79.
- Confirmed the overall quality of the fold, as the score falls strictly within the range of native proteins.



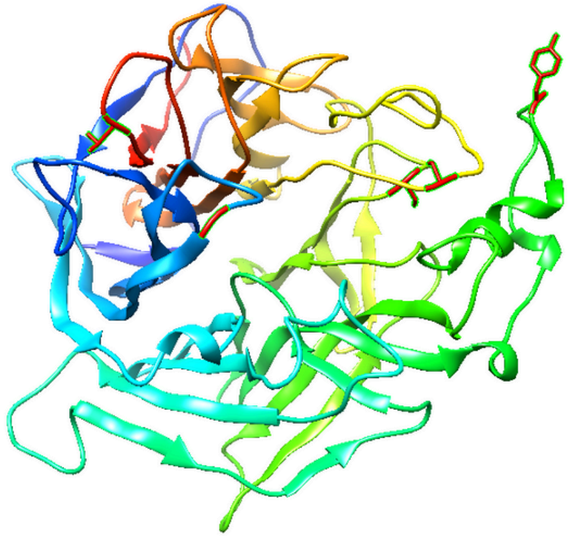
Structural Analysis

- Reveals a classic six-bladed beta-propeller fold, a structural signature typical of the *Henipavirus* attachment glycoproteins.
- Shows a dominance of anti-parallel beta-sheets (highlighted in yellow), which arrange symmetrically to form the "blades" of the propeller.
- Highlights the conserved architecture required for binding to the host receptor EFNB2, suggesting a shared infection mechanism with the *Hendra and Nipah viruses.*

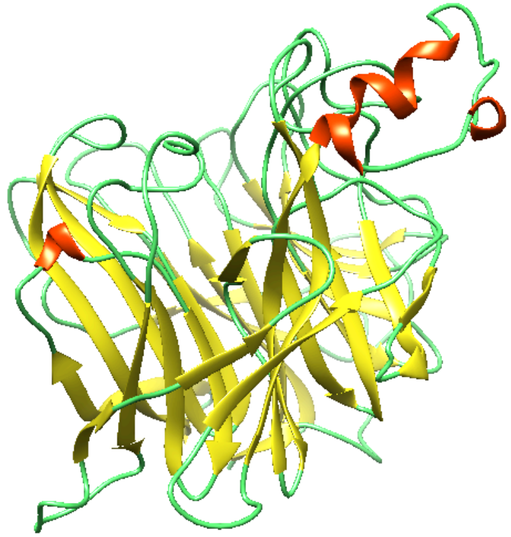
Detailed Structural Views



The solvent-accessible surface area and potential binding pockets.

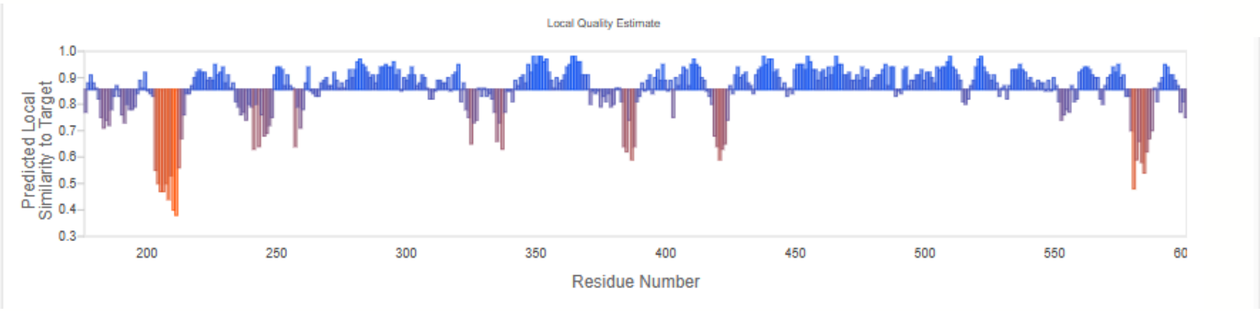


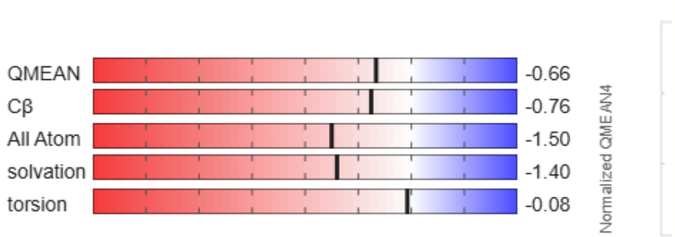
The peptide chain path from the N-terminus (Blue) to the C-terminus (Red).



Highlights the loop regions (green) connecting the beta-strands (yellow).

Local Quality Estimate





- Analyzed the structural reliability of each individual amino acid residue to identify potential stability issues across the protein chain.
- Observed consistently high confidence scores (in blue) that align with the stable beta-sheet regions, confirming the rigidity of the propeller blades.
- Identified that the dips in the graph (in orange) correspond to flexible loops on the protein surface rather than modeling errors.
- Validated the model's biological accuracy, demonstrating a realistic balance between a stable internal core and necessary surface flexibility.

Conclusion

A three-dimensional homology model of the *HeV-g2* attachment glycoprotein was successfully generated using a *Hendra virus*-based structural template. The model demonstrated strong structural reliability, as indicated by superior validation metrics, including a Z-score of −7.91 and 88.4% residues in favoured regions, outperforming the published reference study. Structural analysis confirmed the presence of a conserved six-bladed β-propeller fold, a hallmark feature essential for host receptor recognition and viral attachment.

References

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